

Comparison Operators in SQL Server with Examples

In this article, I am going to discuss the **Comparison Operators in SQL Server** with Examples. Please read our previous article where we discussed the [Arithmetic Operators](#) with examples. At the end of this article, you will understand what are comparison operators and their use in SQL server and different type of comparison operators like equal (=), greater than (>), less than (<), greater than or equal to (>=), less than or equal to (<=), not equal (<>), etc. in detail with examples.

What are Comparison Operators in SQL Server?

The comparison operators in SQL Server are used to compare two values. That means comparison operators are used to comparing one expression with another expression. The result of a comparison can be TRUE, FALSE, or UNKNOWN (an operator that has one or two NULL expressions returns UNKNOWN).

Note: You can use comparison operators on all the expressions except expressions of type text, ntext, or image.

Types of Comparison Operators in SQL Server

The different types of comparison operators that are available in SQL Server are as follows:

1. **Equal (=) Operator**
2. **Not Equal (!= or <>) Operator**
3. **Greater Than (>) Operator**
4. **Less Than (<) Operator**
5. **Greater Than or Equal To (>=) Operator**
6. **Less Than or Equal To (<=) Operator**
7. **Not Greater Than (!<) Operator**
8. **Not Less Than (!>) Operator**

Understanding the Comparison Operators in SQL Server with Examples:

Let us understand how to use the comparison operators with some examples. We are going to use the following Employee table to understand these operators.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Please use the below script to create the database EmployeeDB, Employee table, and populate Employee table with the test data.

```
--Create database EmployeeDB
Create database EmployeeDB
Go
Use EmployeeDB
Go
-- Create Employee table
CREATE TABLE Employee
(
    ID INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(100),
    EmailID VARCHAR(100),
    Gender VARCHAR(100),
    Department VARCHAR(100),
    Salary INT,
    Age INT,
    CITY VARCHAR(100)
)
GO
--Insert some test data into Employee table
INSERT INTO Employee VALUES('Pranaya','Pranaya@g.com','Male','IT',25000,30,
INSERT INTO Employee VALUES('Tarun','Tarun@g.com','Male','Payroll',30000,27,
INSERT INTO Employee VALUES('Priyanka','Priyanka@g.com','Female','IT',27000,25,'Bangalore')
INSERT INTO Employee VALUES('Preety','Preety@g.com','Female','HR',35000,26,'Bangalore')
INSERT INTO Employee VALUES('Ramesh','Ramesh@g.com','Male','IT',26000,27,'M
```

```

INSERT INTO Employee VALUES('Pramod','Pramod@g.com','Male','HR', 29000, 28,'Odisha')
INSERT INTO Employee VALUES('Anurag','Anurag@g.com','Male','Payroll', 27000, 26,'Odisha')
INSERT INTO Employee VALUES('Hina','Hina@g.com','Female','HR', 26000, 30,'Mumbai')
INSERT INTO Employee VALUES('Sambit','Sambit@g.com','Male','Payroll', 30000, 27,'Mumbai')
INSERT INTO Employee VALUES('Manoj','Manoj@g.com','Male','HR', 30000, 28,'Odisha')
INSERT INTO Employee VALUES('Sara','Sara@g.com','Female','Payroll', 28000, 27,'Mumbai')
INSERT INTO Employee VALUES('Lima','Lima@g.com','Female','HR', 30000, 30,'Bangalore')
INSERT INTO Employee VALUES('Dipak','Dipak@g.com','Male','Payroll', 32000, 25,'Bangalore')
GO

```

Equal (=) Operator in SQL Server

The equal (=) operator in SQL Server is used to check whether the two expressions are equal or not. If it's equal then the condition becomes true and it will return the matched records. For example, the below SQL statement will return records from the Employee table where the CITY is Mumbai.

SELECT * FROM Employee WHERE CITY = 'Mumbai'

Once you execute the above query, you will get the following output.

ID	Name	EmailID	Gender	Departm...	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai

Not Equal (!=) Operator in SQL Server

This is the opposite of the equal operator. That means the not equal (!=) operator is used to check whether the two expressions are equal or not. If it's not equal then the condition becomes true and it will return the not-matched records. For example, the following SQL query will return all records from the Employee table except the employees whose CITY is Mumbai.

SELECT * FROM Employee WHERE CITY != 'Mumbai'

Once you execute the above SQL query, you will get the following output.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Not Equal (<>) Operator in SQL Server

The not equal (<>) operator in SQL Server is used to check whether the two expressions are equal or not. If it's not equal then the condition becomes true and it will return the not-matched records. The example is given below.

SELECT * FROM Employee WHERE CITY <> 'Mumbai'

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Note Both != and <> operators are not equal operators and will return the same result but != operator is not an ISO standard.

Greater Than (>) Operator in SQL Server

The Greater Than (>) operator in SQL Server is used to check whether the left-hand operator is **higher than** the right-hand operator or not. If the left-hand operator **higher than** the right-hand operator then the condition becomes true and it will return the matched records. The following SQL query will return records from the Employee table where the ID is greater than 5.

SELECT * FROM Employee WHERE ID > 5

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Less Than (<) Operator in SQL Server

The less than (<) operator in SQL Server is used to check whether the left-hand operator is **lower than** the right-hand operator or not. If the left-hand operator **lower than** the right-hand operator then the condition becomes true and it will return the matched records. The below SQL statement will return records from the Employee table where the ID is less than 5.

SELECT * FROM Employee WHERE ID < 5

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore

Greater Than or Equal To (>=) Operator in SQL Server

The Greater than or Equal To (>=) operator in SQL Server is used to check whether the left-hand operator is **higher than or equals to** the right-hand operator or not. If the left-hand operator is **higher than or equals to** the right-hand operator then the condition becomes true and it will return the matched records. The following SQL statement will return records from the Employee table where the Salary is **greater than or equal to 29000**.

SELECT * FROM Employee WHERE Salary >= 29000

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

Less Than or Equal To (<=) Operator in SQL Server

The Less Than or Equal To (<=) operator in SQL Server is used to check whether the left-hand operator is lower than or equals to the right-hand operator or not. If the left-hand operator is lower than or equals to the right-hand operator then the condition becomes true and it will return the matched records. The below SQL query will return records from the Employee table where the Salary is **less than or equal to 29000**.

SELECT * FROM Employee WHERE Salary <= 29000

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai

Not Greater Than (!>) Operator in SQL Server

The Not Greater Than (!>) operator in SQL Server is used to check whether the left-hand operator is not greater than the right-hand operator or not. If the left-hand operator is not greater than the right-hand operator then the condition becomes true and it will return the matched records. The following SQL statement will return records from the Employee table where the employee ID is **not greater than 5**.

SELECT * FROM Employee WHERE ID !> 5

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Departm...	Salary	Age	CITY
1	Pranaya	Pranaya@g.com	Male	IT	25000	30	Mumbai
2	Tarun	Tarun@g.com	Male	Payroll	30000	27	Odisha
3	Priyanka	Priyanka@g.com	Female	IT	27000	25	Bangalore
4	Preety	Preety@g.com	Female	HR	35000	26	Bangalore
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai

Not Less Than (!<) Operator in SQL Server

The Not Less Than (!<) operator in SQL Server is used to check whether the left-hand operator is not less than the right-hand operator or not. If the left-hand operator is not less than the right-hand operator then the condition becomes true and it will return the matched records. The below SQL query will return records from the Employee table where the employee ID is **not less than 5**.

SELECT * FROM Employee WHERE ID !< 5

Following is the output of the above SQL query.

ID	Name	EmailID	Gender	Department	Salary	Age	CITY
5	Ramesh	Ramesh@g.com	Male	IT	26000	27	Mumbai
6	Pramod	Pramod@g.com	Male	HR	29000	28	Odisha
7	Anurag	Anurag@g.com	Male	Payroll	27000	26	Odisha
8	Hina	Hina@g.com	Female	HR	26000	30	Mumbai
9	Sambit	Sambit@g.com	Male	Payroll	30000	25	Odisha
10	Manoj	Manoj@g.com	Male	HR	30000	28	Odisha
11	Sara	Sara@g.com	Female	Payroll	28000	27	Mumbai
12	Lima	Lima@g.com	Female	HR	30000	30	Bangalore
13	Dipak	Dipak@g.com	Male	Payroll	32000	25	Bangalore

In the next article, I am going to discuss the [Logical Operators in SQL Server](#). Here, in this article, I try to explain the **Comparison Operator in SQL Server** with Examples. I hope this article will help you with your needs. I would like to have your feedback. Please post your feedback, question, or comments about this **Comparison Operator in SQL Server** with Examples article.